ACS COLLEGE OF ENGINEERING, BENGALURU
DEPARTMENT OF AEROSPACE ENGINEERING
Affiliated to Visvesvaraya Technological University (VTU), Belagavi | Approved by AICTE, New Delhi
Bachelor of Engineering (B.E.) in Aerospace Engineering — Final Year Major Project (Phase-II) • Academic Year 2025–2026 / 2026–2027

Precise Trajectory Propagation and Events Prediction for Highly Eccentric Orbits

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ABSTRACT

Spacecraft in Highly Eccentric Orbits (HEOs)—Geostationary Transfer Orbits (GTO), critical-inclination Molniya orbits ($i \approx 63.4°$), Tundra and high-apogee science orbits—pass within each revolution from a fast, strongly perturbed perigee (a few hundred kilometres up, above 10 km/s), where atmospheric drag and the Earth's oblateness dominate, to a slow apogee dwell beyond 35,000 km, where the Sun and Moon dominate. Propagating them accurately requires a complete force model, step control that follows these changes, and care at the discontinuities of the models themselves, such as the edges of the Earth's shadow. This project presents AstroFlyer HEO, an offline desktop application for precise trajectory propagation and event prediction in such orbits. Its engine, written in Python with NumPy and SciPy, integrates the equations of motion in Cowell form in the GCRF frame with EGM2008 gravity up to degree and order 70, JPL DE440 Sun and Moon, solar radiation pressure with a conical shadow cast by the WGS-84 ellipsoid, NRLMSISE-00 atmospheric drag and post-Newtonian relativity, using IAU 2006/2000A Earth orientation with IERS data. The default integrator, Dormand–Prince DOP853, is stopped and restarted exactly at every shadow edge and atmospheric-model jump so that step control never straddles a discontinuity; Radau, RKF7(8), Bulirsch–Stoer, SGP4 and analytic Kepler and J2 propagators are also provided. Apsides, nodes, eclipses and ground-station contacts are found by root-finding during integration. Two propagation strategies were built and measured: covariance-steered step control, which limits the growth of the linearised position uncertainty per step, and multi-fidelity gravity tiers with a certified bound on the omitted acceleration. The propagator was cross-checked against three independent tools—Orekit, NASA GMAT and Basilisk—from the same initial states on Molniya, Tundra, GTO and a stress orbit ($e = 0.88$), adding force models one at a time. It agrees with Orekit to within 0.63 m after one day and 6.1 m after seven days with the full model, and with GMAT to within 1.06 m and 32.6 m; with gravity, Sun and Moon alone the agreement is a few centimetres per day. Every larger difference was traced by experiment to a documented modelling choice, such as a spherical rather than ellipsoidal Earth shadow in GMAT and Basilisk. The application runs entirely offline, with a three-dimensional orbit view, ground track, parameter plots, mission scripting and CCSDS data exchange.

Index Terms / Keywords— *Highly Eccentric Orbits (HEOs), Trajectory Propagation, Cowell Formulation, Dormand–Prince Integration, Force Modelling, Eclipse Prediction, Event Detection, Cross-Validation, Orekit, NASA GMAT, Basilisk.*

Core Focus: Highly Eccentric Orbits (HEOs)	**Formulation:** Cowell, GCRF, EGM2008 70×70 / DE440 / NRLMSISE-00
Engine: Python, NumPy, SciPy (Numba optional)	**Integrators:** DOP853 (default), Radau, RKF7(8), Bulirsch–Stoer
Accuracy vs Orekit: ≤ 0.63 m (1 day), ≤ 6.1 m (7 days)	**Cross-checks:** Orekit • NASA GMAT • Basilisk
Execution Mode: 100% Offline Standalone Desktop App	**Console:** Three.js 3D view, ground track, plots (WebView2)